

**Cambridge (CIE) IGCSE Maths
Paper 2: Non-Calculator (Set B)
Extended**

Saturday 26 April 2025

Morning (Time: 2 hours 0 minutes)

Total marks

/ 100

Instructions

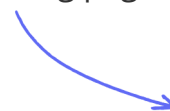
- Try to complete this mock exam paper in one sitting, under exam conditions. Use all the time available and check your answers to each question at the end before submitting.
- Remember this is PRACTICE. Mistakes are fine and will help you improve in time for the real exam - just do your best.
- You will need geometrical instruments.
- Calculators must not be used in this paper.

Materials

- [List of formulas](#)

Scan here to mark your mock exam

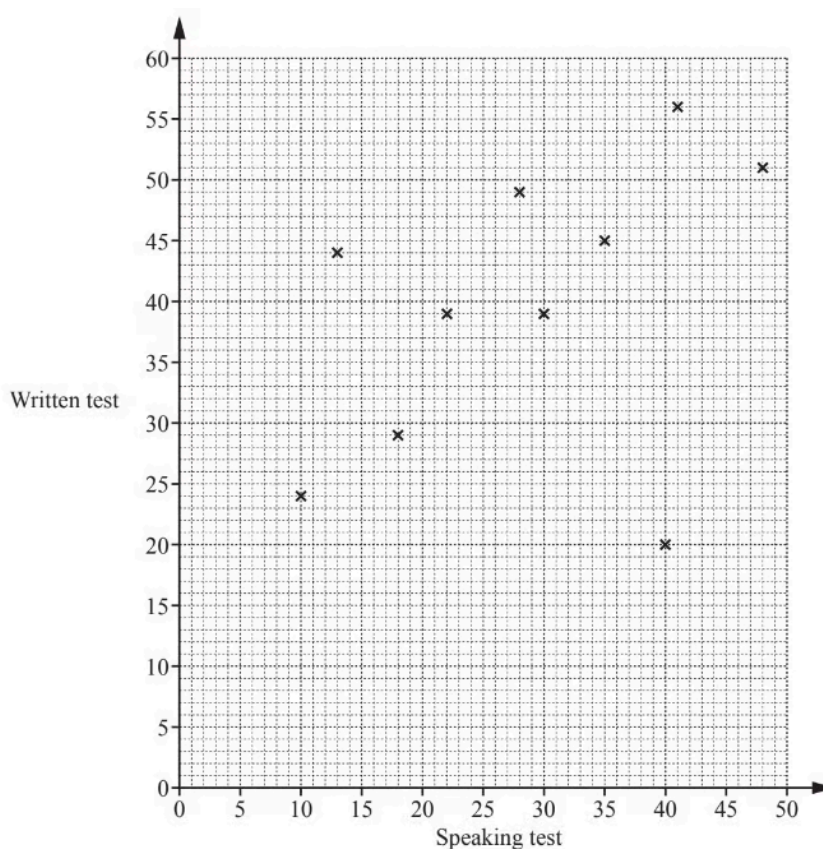
or visit the mock exams landing page for this course



- 1 (a)** Fourteen students each take two tests in French, a speaking test and a written test. The table shows the scores.

Speaking test	10	13	48	30	35	18	41	40	22	28	20	44	37	46
Written test	24	44	51	39	45	29	56	20	39	49	33	52	44	52

Complete the scatter diagram. The first ten points have been plotted for you.



(2 marks)

- (b)** What type of correlation is shown in this scatter diagram?

(1 mark)

- (c)** One student has a high score in the speaking test and a low score in the written test.

On the scatter diagram, put a ring around this point.

(1 mark)

- (d)** On the scatter diagram, draw a line of best fit.

(1 mark)

- (e)** Use your line of best fit to estimate a score in the written test for a student who scored 25 in the speaking test.

(1 mark)

2 (a)

27	14	8	93	32	55	14	38	73	47
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From this list of numbers, find the median.

[2]

(2 marks)

(b)

27	14	8	93	32	55	14	38	73	47
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From this list of numbers, find the range.

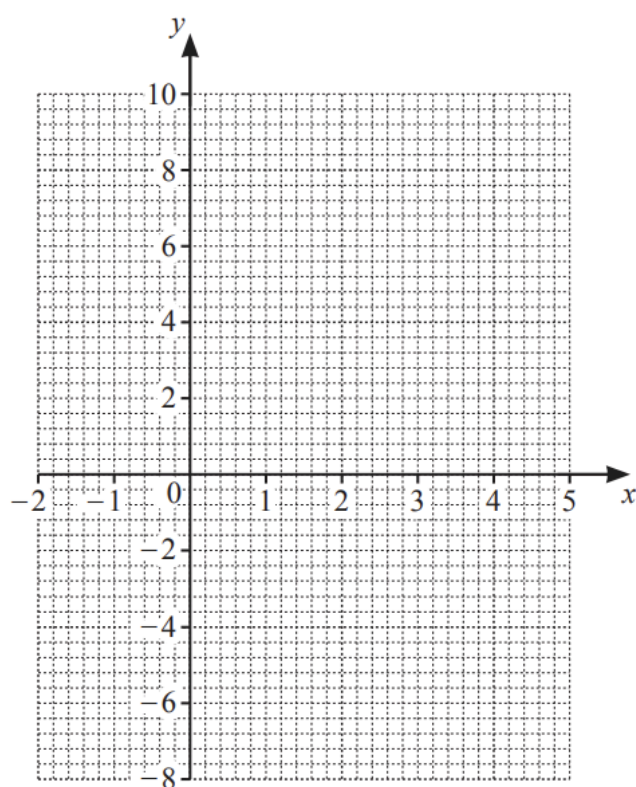
(1 mark)

3 (a) Complete the table of values for $y = x^2 - 4x - 3$

x	-2	-1	0	1	2	3	4	5
y		2	-3	-6		-6	-3	2

(2 marks)

(b) On the grid, draw the graph of $y = x^2 - 4x - 3$ for $-2 \leq x \leq 5$.



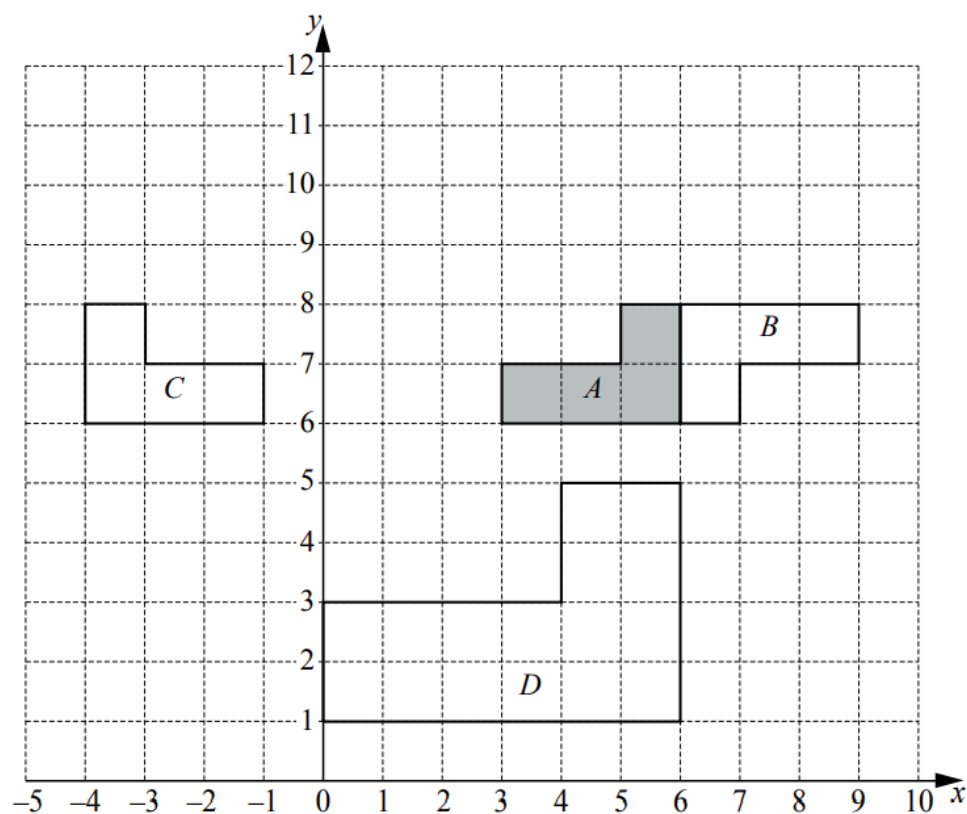
(4 marks)

(c) Use your graph to solve the equation $x^2 - 4x - 3 = 0$.

$x = \dots\dots\dots$ or $x = \dots\dots\dots$

(2 marks)

4



The diagram shows four shapes A , B , C and D .

Describe fully the **single** transformation that maps shape A onto

i) shape B ,

[3]

ii) shape C ,

[2]

iii) shape D .

[3]

(8 marks)

5 Make m the subject of the formula $y = 4m - p$

$m = \dots\dots\dots$

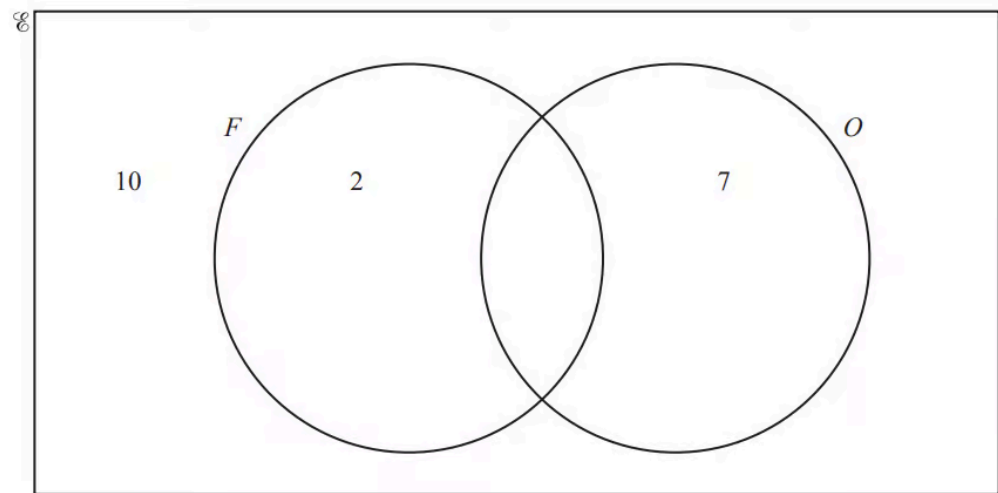
(2 marks)

6 $\mathcal{E} = \{x : x \text{ is a natural number } \leq 15\}$

$F = \{x : x \text{ is a factor of } 12\}$

$O = \{x : x \text{ is an odd number}\}$

i) Complete the Venn diagram to show the elements of these sets.



[2]

ii) Write down one number that is in set O , but not in set F .

[1]

iii) Find $n(F \cup O)$.

[1]

iv) A number is chosen at random from $\mathcal{E}$.

Work out the probability that this number is in set O .

[1]

(5 marks)

7 The length of one side of a rectangle is 12cm.

The length of the diagonal of the rectangle is 13cm.

Calculate the area of the rectangle.

..... cm²

(3 marks)

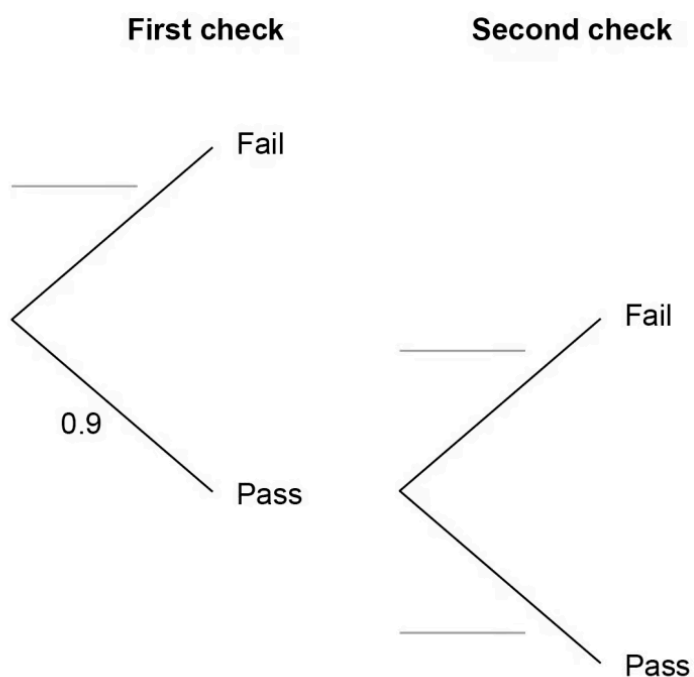
8 (a) Items made at a factory have to pass two checks.

90% pass the first check.

The items that fail are scrapped. 99% of the items that pass the first check pass the second check.

The items that fail are scrapped.

Complete the tree diagram.



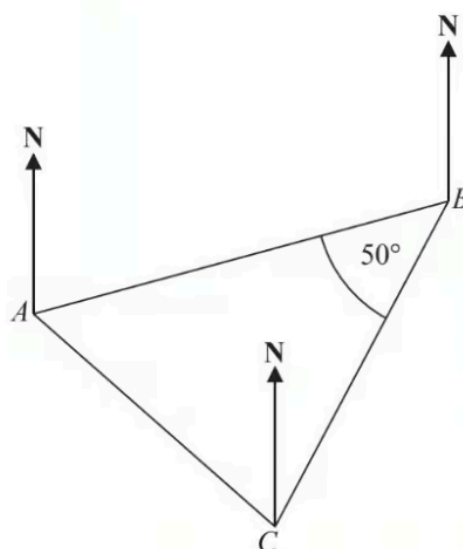
(2 marks)

(b) An item is chosen at random before the checks.

Work out the probability that the item is scrapped.

(3 marks)

- 9 The diagram shows the positions of three points, A , B and C , on a map.



The bearing of B from A is 070° .

Angle ABC is 50° .

$AB = CB$

Work out the bearing of C from A .

(3 marks)

- 10** i) 29 parcels each have a value of \$68. By writing each of these numbers correct to 1 significant figure, find an estimate for the total value of these 29 parcels.

\$ [1]

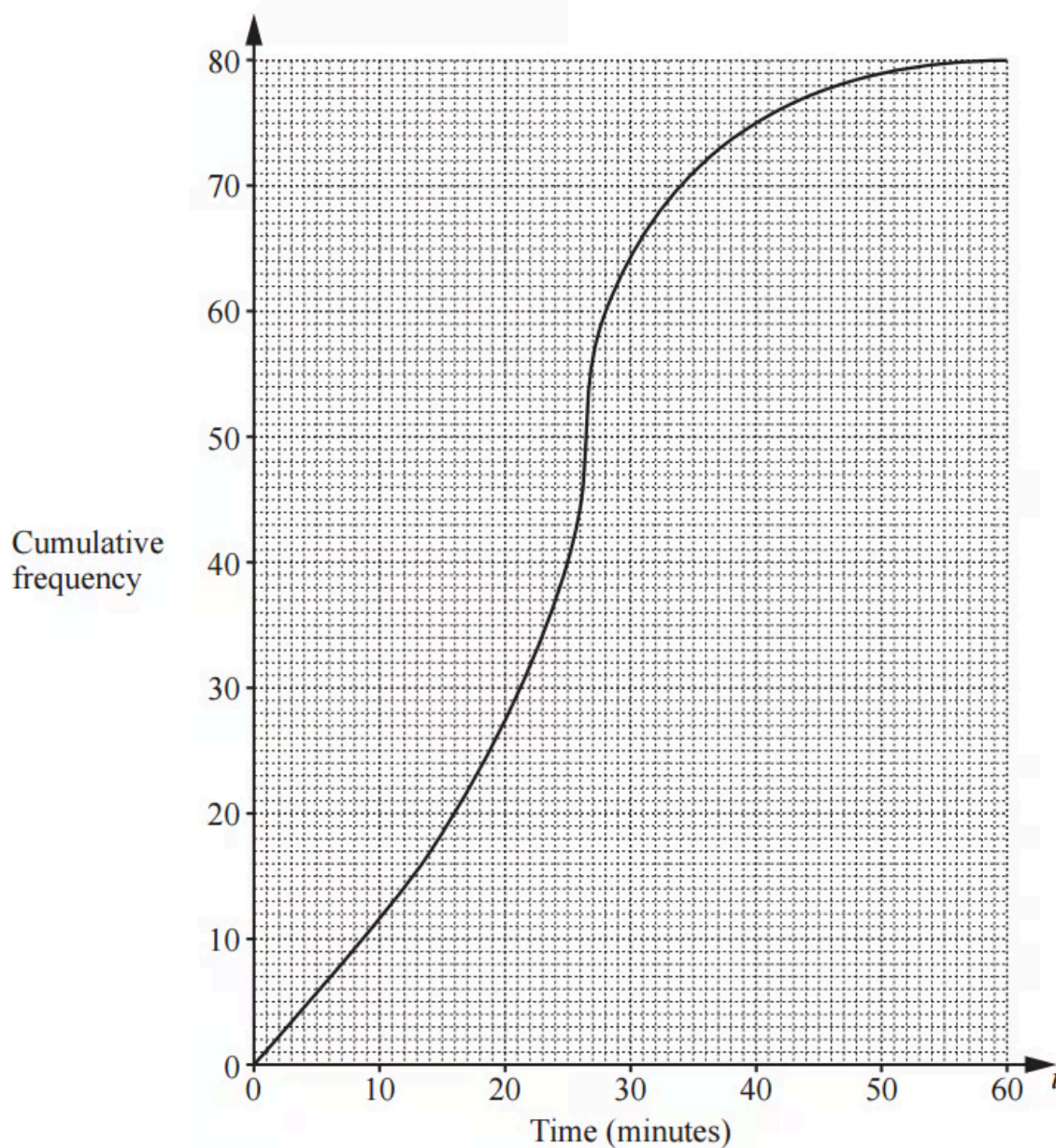
- ii) Without doing any calculation, complete this statement.

The actual total value of these 29 parcels is less than the answer to **part (i)**

because [1]

(2 marks)

- 11 (a)** The time, t minutes, 80 students each spend completing their homework is recorded. The cumulative frequency diagram shows the results.



Use the cumulative frequency diagram to find an estimate of the median.

..... min

(1 mark)

(b) Use the cumulative frequency diagram to find an estimate of the interquartile range.

..... min

(2 marks)

(c) Use the cumulative frequency diagram to find an estimate of the number of students who spend more than 40 minutes completing their homework.

(2 marks)

12 Show that $\frac{\sqrt{150} - \sqrt{6}}{\sqrt{2} \times \sqrt{3}}$ simplifies to an integer.

(3 marks)

13 (a) The n th term of a sequence is given by $an^2 + bn$ where a and b are integers.

The 2nd term of the sequence is -2

The 4th term of the sequence is 12

Find the 6th term of the sequence.

(4 marks)

(b) Here are the first five terms of a different quadratic sequence.

0 2 6 12 20

Find an expression, in terms of n , for the n th term of this sequence.

(2 marks)

14

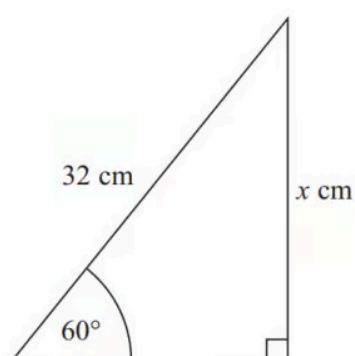


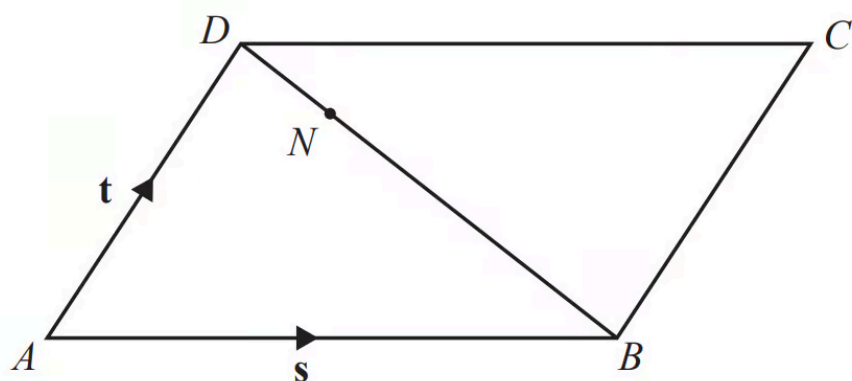
Diagram **NOT**
accurately drawn

Calculate the value of x .

Give your answer as an exact value.

(3 marks)

15 (a)



NOT TO
SCALE

$ABCD$ is a parallelogram.

N is the point on BD such that $BN:ND = 4:1$.

$\vec{AB} = \mathbf{s}$ and $\vec{AD} = \mathbf{t}$.

Find, in terms of $\mathbf{s}$ and $\mathbf{t}$, an expression in its simplest form for $\vec{BD}$.

$\vec{BD} = \dots\dots\dots$

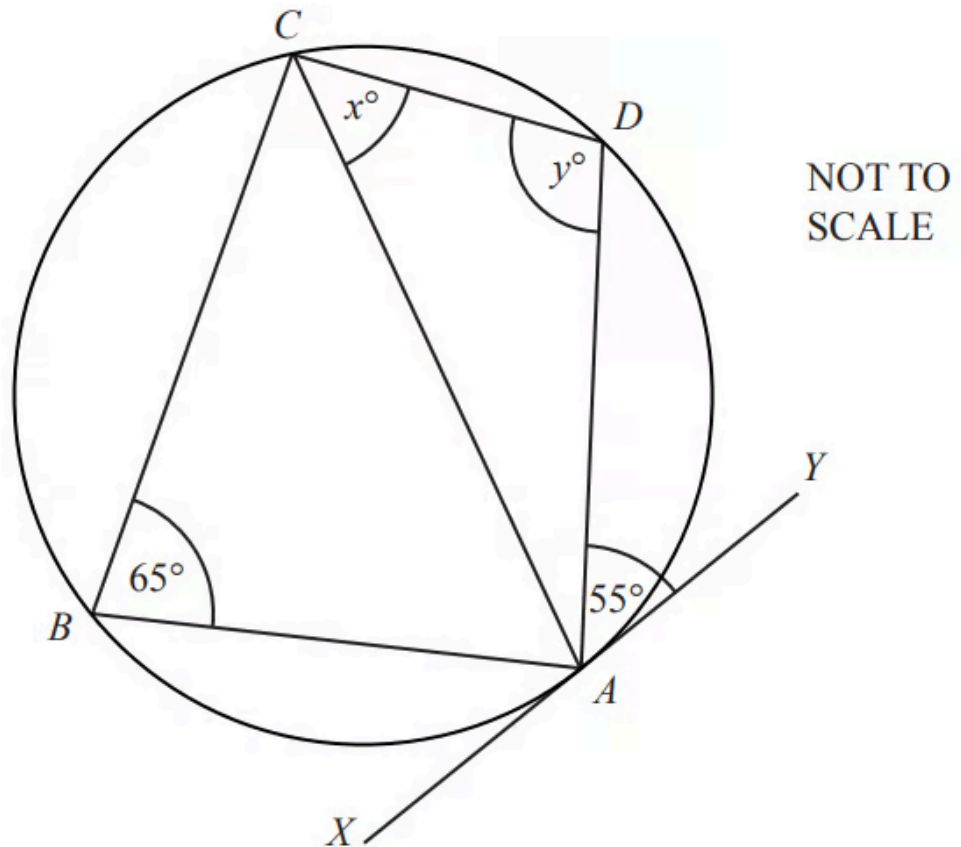
(1 mark)

(b) Find, in terms of $\mathbf{s}$ and $\mathbf{t}$, an expression in its simplest form for $\vec{CN}$.

$\vec{CN} = \dots\dots\dots$

(3 marks)

16 (a)



A, B, C and D are points on the circumference of the circle.
The line XY is a tangent to the circle at A .

Find the value of x , giving a reason for your answer.

$x = \dots\dots\dots$ because $\dots\dots\dots$

(2 marks)

(b) Find the value of y , giving a reason for your answer.

$y = \dots\dots\dots$ because $\dots\dots\dots$

(2 marks)

17 Work out $2\frac{2}{3} + 3\frac{1}{2}$

Give your answer as a mixed number in its simplest form.

(3 marks)

18

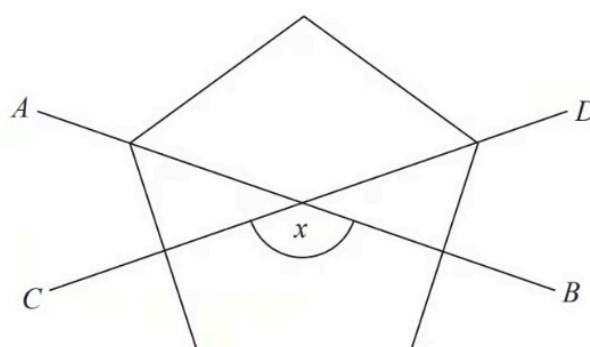


Diagram **NOT**
accurately drawn

The diagram shows a regular pentagon.

AB and *CD* are two of the lines of symmetry of the pentagon.

Work out the size of the angle marked *x*.

You must show all your working.

(4 marks)

19 Simplify $8^2 \times \sqrt[3]{4^6}$

Give your answer in the form 2^a where a is an integer.

Show each stage of your working clearly.

(3 marks)

20

$$\frac{3}{m+4} - \frac{4}{m} = 6$$

Show that this equation can be written as $6m^2 + 25m + 16 = 0$.

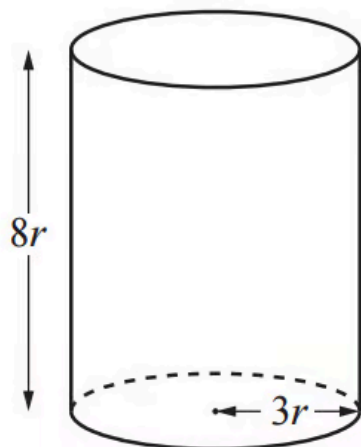
(3 marks)

21 $f(x) = \frac{1}{2}x$ $g(x) = x - x^2$

Solve $f^{-1}(x) = gf(x)$

(4 marks)

22 (a)



NOT TO
SCALE

The diagram shows a solid cylinder and a solid sphere.

The cylinder has radius $3r$ and height $8r$.

The sphere has radius r .

Find the volume of the sphere as a fraction of the volume of the cylinder.

Give your answer in its lowest terms.

[The volume, V , of a sphere with radius r is $V = \frac{4}{3} \pi r^3$.]

(4 marks)

(b) The surface area of the sphere is $81\pi \text{ cm}^2$.

Find the **curved** surface area of the cylinder.

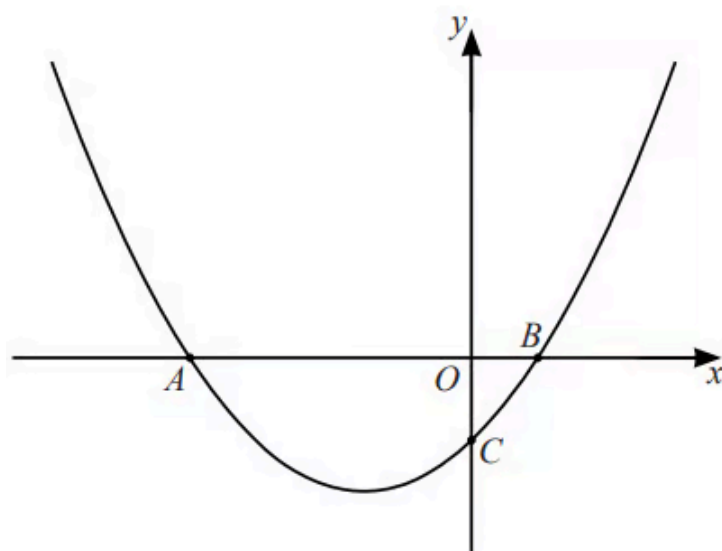
Give your answer in terms of π .

[The surface area, A , of a sphere with radius r is $A = 4\pi r^2$.]

..... cm^2

(4 marks)

23



NOT TO
SCALE

The diagram shows a sketch of the curve $y = x^2 + 3x - 4$.

i) Differentiate $y = x^2 + 3x - 4$

[2]

ii) Find the equation of the tangent to the curve at the point (2, 6).

[3]

(5 marks)